

Modeling 3D Hair by Outlining Hair Cards

Masamune Takano

Kyushu University, Japan

takano.masamune.381@s.kyushu-u.ac.jp

Yuki Morimoto

Kyushu University, Japan

morimoto@design.kyushu-u.ac.jp

ACM Reference Format:

Masamune Takano and Yuki Morimoto. 2022. Modeling 3D Hair by Outlining Hair Cards. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '22 Posters)*, August 07–11, 2022, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532719.3543214>

1 INTRODUCTION

We propose an intuitive method for modeling 3DCG character's hair by inputting sketched lines as the left and right contour lines of a hair strand. Modeling a character's hairstyle is an important task that affects the character's appearance. Generally, hair modeling requires many complex tasks. In our method, the user inputs the left and right outlines by sketching them to create hair strand meshes. This enables the user to adjust the details while considering the overall outline of the hairstyle (Fig. 1).

2 RELATED WORKS

In order to solve the problem that detailed hair modeling requires many complex tasks, some methods using sketches or parametric curves have been proposed.

In Hairbrush [Xing et al., 2019], users can model hairstyles using a VR interface and a trained database that contains data about the shape of hairstyle models. The user draws lines on a head model to indicate the direction of hair growth using a VR controller, and the database analyses the lines and suggests hairstyles. Then the user chooses one hairstyle from the suggestions and generates a part of the hair from the chosen hairstyle. However, HairBrush is not appropriate for creating new hairstyles because it creates hairstyles by merging existing hairstyles from its database.

In DeepSketchHair [Shen et al., 2021], the user can create hair models by drawing lines on a head model to indicate a hair contour or the direction of hair growth. Real-time rendering is difficult because the hair is a photorealistic and high-poly model that describes every strand of hair. Additionally, it is difficult to adjust the details of the hair because the entire model is made at once.

The ability to create a hair model by extruding the mesh from the NURBS or Bézier curve is installed in many types of 3DCG software, for example, Autodesk Maya, Blender, and CINEMA4D. The user inputs two curves and sweep one along the other. The trail of the curve's movement is converted into a 3D mesh. In this method, detail adjustment and UV mapping are required for each hair strand.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Posters, August 07–11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9361-4/22/08.

<https://doi.org/10.1145/3532719.3543214>

In AnimeHairSupporter [relarar, 2019], which is an add-on of Blender, the user draws a NURBS or Bézier curve and uses it as an axis to form hair strands. However, it is difficult to adjust the left and right sides of the hair strands using AnimeHairSupporter.

3 OUR METHOD

In our method, we define a template as a canvas and place it around a head model. The user draws on the template to model the hair; the user intuitively provides the entire silhouette by drawing lines as the left and right sides of the contour of the hair cards (Sec. 3.3). The modeling workflow is as follows:

1. Sketch two lines on the head model and template (Fig. 2a).
2. Simplify the lines (Fig. 2b).
3. Generate thick, round hair card geometry (Fig. 2c).
4. Generate the mesh for UV mapping (Fig. 2d).

The details of this workflow are shown in Sec. 3.1 and later.

3.1 Draw two lines on the template

The user draws a line by dragging the mouse to create the side of a hair card. The starting point is placed on the intersection of the ray, which is cast from the viewpoint, and the head model. Similarly, subsequent points are placed where the ray intersects the template, as the starting point. When the user inputs two lines, these are resampled [Shin-Ting et al., 2004]. Then, these vertices are smoothed (Fig. 3 left).

3.2 Create the geometry of a hair card

To add thickness to the hair cards, we calculate the vertices v (the green points in Fig. 3c) between each pair of vertices a_i and b_i as follows:

$$v_{i,j} = \frac{j}{s} \vec{a_i b_i} + j(j-s) \vec{r_i n_i}, \quad r_i = \begin{cases} r_0 \left(\frac{j}{m} \right) & (i < m) \\ r_0 & (\text{otherwise}) \end{cases} \quad (1)$$

Where n_i is the normal vector of the mesh that consists of a_i , a_{i+1} , and b_i ; s is the number of points used to divide the card in direction $\vec{a_i b_i}$; i is the vertex number; r_0 is a positive real number that represents the initial value of r_i , and m is an adjustable natural number that represents the gradual rise of hair from the surface of the head model to the template. For our implementation, $r_0 = 0.09$, $m = 4$, $n = 10$, $s = 10$. Based on a_i , b_i , and v , we generate a polygon mesh for UV texture mapping. By repeating this procedure from drawing to texture mapping, our system enables the user to generate the detailed parts of a hairstyle while considering the overall outline of the hairstyle.

3.3 Our template model

Three types of templates are available in our method as follows. Note that we apply the geometric modeling of a card described in

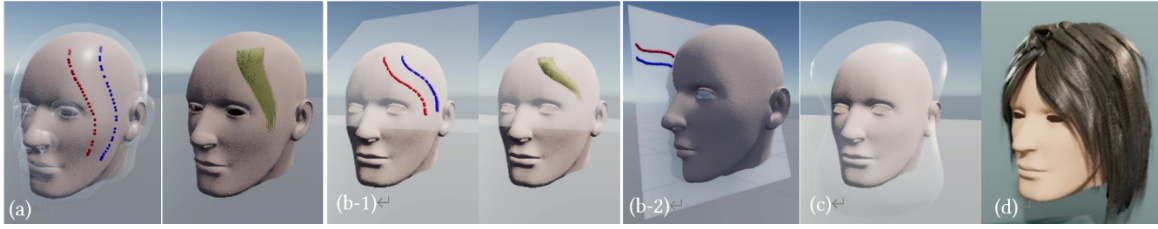


Figure 1: Generate hair cards by inputting sketch lines on our three types of templates, finally create various hairstyles.

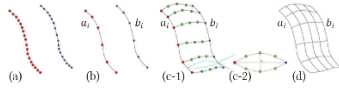


Figure 2: Geometry of a hair card: (a) sketched lines, (b) simplify, (c-1) insert the intermediate vertices, (c-2) sectional view, and (d) generate the mesh.

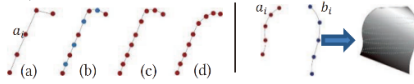


Figure 3: and smoothing (left), and Making asymmetrical template by drawing two lines (right).

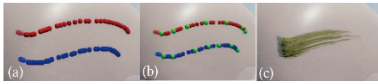


Figure 4: (a) Input two lines, (b) simplify (the green points), and (c) generate meshes for texture mapping.

Section 3.2 to the templates using OverCoat [Schmid et al., 2011] and an arbitrary shape.

4 OVERCOAT TEMPLATE:

We apply OverCoat to create slightly enlarged models to cover and fit any head model (Fig. 1a). While using the templates based on OverCoat, we represent that only the roots of human hair are buried in the scalp, whereas the remainder is separate.

5 PLANE TEMPLATE:

There are two types of plane template: one creates a tangent surface at the location the user clicks using the mouse (Fig. 1b-1), whereas the other creates a plane centered at the same coordinates as the center of the head model while the plane is perpendicular to the ray from the viewpoint (Fig. 1b-2). The user can switch these plane templates to any type of template while creating a hairstyle. These templates enable the user to create upside-down or partly bouncy hairstyles, and these are useful for making jumped-up hairs.

The plane templates appear when the user draws the first point so that the first point is always placed on the surface of the head. Then the card is generated by the user drawing two lines (3.2).

6 ARBITRARILY SHAPED TEMPLATE:

Other arbitrary shapes can be used as templates (Figs. 1c) by loading 3D models. The user can switch the template to any template while creating a hairstyle. These templates enable the user to create upside-down or partly bouncy hairstyles and generate hairstyles with various silhouettes. Furthermore, the user can create an asymmetrical template by drawing two lines around the head model that give the left and right silhouettes (Fig. 3 right)

7 RESULTS AND CONCLUSION

The results of our method are shown in Fig. 1d. For the system runtime environment, the CPU was AMD Ryzen 9 3900X, GPU was GeForce RTX 2080 Ti, and the system was operated interactively. Whereas general methods take hours to modeling, this method took about 20~30 minutes for making a hairstyle.

We succeeded in modeling hairstyles with a variety of silhouettes using our hairstyle templates. We proposed a method for hair modeling that uses drawing lines as the outline of a hair card. We demonstrated that our method enables the user to create a variety of hairstyles. In future work, we would like to animate hair cards, in addition to hair modeling, so that the generated cards sway under external forces, like real hair. Additionally, we will improve the functionality, for example, by improving the UI.

REFERENCES

- Jun Xing, Koki Nagano, Weikai Chen, Haotian Xu, Li-yi Wei, Yajie Zhao, Jingwan Lu, Byungmoon Kim, and Hao Li, 2019. HairBrush for Immersive Data-Driven Hair Modeling. In Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology. Association for Computing Machinery, New York, NY, USA, 263–279. DOI: <https://doi.org/10.1145/3332165.3347876>
- Yuefan Shen, Changgeng Zhang, Hongbo Fu, Kun Zhou, and Youyi Zheng, 2021. DeepSketchHair: Deep Sketch-Based 3D Hair Modeling. IEEE Transactions on Visualization and Computer Graphics 27, 7, 3250–3263. DOI: <https://doi.org/10.1109/TVCG.2020.2968433>
- relarar, 2019. <https://github.com/relarar/Blender-AnimeHairSupporter>
- Wu Shin-Ting, Adler C. G. da Silva, and Mercedes Rocio Gonzales Márquez, 2004. The Douglas-Peucker Algorithm: Sufficiency Conditions for Non-Self-Intersections. J. Braz. Comput. Soc., 9, 67–84.
- Johannes Schmid, Martin Sebastian Senn, Markus Gross, Robert W. Sumner, 2011. OverCoat: an implicit canvas for 3D painting. ACM Trans. Graph. 30, 4, Article 28 (July 2011), 10 pages. DOI: <https://doi.org/10.1145/2010324.1964923>